

90

EXAM 1

1) CALC Q OF UN FUEL:

GIVEN: $\epsilon: 19\%$, $\rho_{UN}: 14.3 \text{ g/cm}^3$, $\sigma_{F_{UN}}: 587 \times 10^{-24} \text{ cm}^2$
 $\phi_n: 4 \times 10^{12} \text{ n/cm}^2 \cdot \text{s}$ $M_N = 14 \text{ g/mol}$

$$Q = \epsilon_f \cdot N_f \cdot \sigma_f \cdot \phi_n$$

ASSUME ϵ_f IS $200 \times 10^6 \text{ eV}$

$$M_{UN} = M_{H_4O} + M_{UN} = (0.19(235) + 0.81(238)) = 237.43 \text{ g/mol}$$

$$\Rightarrow M_{UN} = 251.43 \text{ g/mol}$$

$$N_f = \rho_{UN} \cdot M_{UN} \cdot 6.022 \times 10^{23} \cdot \frac{U}{UN} \cdot \epsilon$$

$$N_f = \frac{14.3 \text{ g}}{\text{cm}^3} \cdot \frac{251.43 \text{ g}}{\text{mol}} \cdot \frac{6.022 \times 10^{23}}{\text{mol}} \cdot \frac{10}{UN} \cdot 0.19$$

$$\sqrt{N_f} = 6.507 \times 10^{21} \text{ Ue/s/cm}^3$$

$$Q = 200 \times 10^6 \text{ eV} (1.602 \times 10^{-19} \text{ J/eV}) \cdot (6.507 \times 10^{21} \text{ Ue/s/cm}^3) \cdot (4 \times 10^{12} \text{ n/cm}^2 \cdot \text{s}) \cdot (587 \times 10^{-24} \text{ cm}^2)$$

$$Q = 489.5 \text{ W/cm}^3$$

FIND ϵ OF UO_2 FOR SAME Q

ASSUME $Q = 489.5 \text{ W/cm}^3$ $\rho_{UO_2}: 10.97 \text{ g/cm}^3$

$$N_f = \rho_{UO_2} \cdot M_{UO_2} \cdot 6.022 \times 10^{23} \cdot \frac{U}{UO_2} \cdot \epsilon$$

$$M_{UO_2} = M_{H_4O} + M_{UN}$$

$$\Rightarrow x(235) + (1-x)(238) + 2(16)$$

$$\Rightarrow 235x + 238 - 238x + 32$$

$$M_{UO_2} = 238 - 3x + 32 = 270 - 3x$$

$$N_f = \frac{10.97 \text{ g/cm}^3}{270 - 3x_{\text{eff}}} \cdot \frac{6.022 \times 10^{23}}{\text{mol}} \cdot x_{\text{eff}}$$

$$6.507 \times 10^{21} \text{ Ue/s/cm}^3 = \frac{10.97}{270 - 3x_{\text{eff}}} (6.022 \times 10^{23} x_{\text{eff}})$$

$$6.507 \times 10^{21} (270 - 3x_{\text{eff}}) = (10.97)(6.022 \times 10^{23})(x_{\text{eff}})$$

$$x_{\text{eff}} = 26.51\% \text{ FOR } UO_2$$

EXAM 1

2. CALCULATE T_c RADIAL MID $\phi r = 0.1 \text{ cm}$

ASSUME GAP IS ALL HC.

$$k_{\text{cnd}} = 0.18 \text{ W/cm} \cdot \text{K}$$

$$h_{\text{cool}} = 1.5 \text{ W/cm}^2 \cdot \text{K}$$

$$k_{\text{fuel}} = 0.04 \text{ W/cm} \cdot \text{K}$$

$$Q = 350 \text{ W/cm}^3$$

$$R_f = 0.45 \text{ cm}$$

$$t_g = 80 \mu\text{m} = 0.008 \text{ cm}$$

$$T_{\text{cool}} = 540 \text{ K}$$

$$b_{\text{cnd}} = 0.06 \text{ cm}$$

$$k_{\text{gap}} = 16 \times 10^{-6} \times T^{0.79}$$

$$* \text{LHR} = \pi R_f^2 Q = \pi \cdot (0.45 \text{ cm})^2 \cdot 350 \text{ W/cm}^3$$

$$\text{LHR} = 222.66 \text{ W/cm}$$

$$* \frac{\text{LHR}}{2\pi R_f} = \frac{222.66 \text{ W/cm}}{2\pi (0.45 \text{ cm})} = 78.75 \text{ W/cm}^2$$

$$T_{\text{co}} - T_{\text{cool}} = \frac{\text{LHR}}{2\pi R_f} \cdot \frac{1}{h_{\text{cool}}} = (78.75 \text{ W/cm}^2 \cdot \frac{1}{1.5 \text{ W/cm}^2 \cdot \text{K}}) + 540 \text{ K}$$

$$T_{\text{co}} = 592.5 \text{ K}$$

$$T_{\text{ci}} - T_{\text{co}} = \frac{\text{LHR}}{2\pi R_f} \cdot \frac{b_{\text{cnd}}}{k_{\text{cnd}}} = 78.75 \text{ W/cm}^2 \cdot \left(\frac{0.06 \text{ cm}}{0.18 \text{ W/cm} \cdot \text{K}} \right) + 592.5 \text{ K}$$

$$T_{\text{ci}} = 618.75 \text{ K}$$

$$T_s - T_{\text{ci}} = \frac{\text{LHR}}{2\pi R_f} \cdot \frac{t_g}{k_{\text{gap}}} \quad h_{\text{gap, cool}} = 16 \times 10^{-6} \cdot (618.75 \text{ K})^{0.79}$$

$$= (78.75 \cdot \frac{0.008 \text{ cm}}{0.0026 \text{ W/cm} \cdot \text{K}}) + 618.75 \text{ K}$$

$$T_s = 861.06 \text{ K}$$

$$T_o - T_s = \frac{\text{LHR}}{4\pi k_f} + 861.06 \text{ K} = \frac{222.66 \text{ W/cm}}{4\pi \cdot 0.04 \text{ W/cm} \cdot \text{K}} + 861.06 \text{ K}$$

$$T_o = 1304.03 \text{ K}$$

IF $T_{\text{ci}} = 618.75 \text{ K}$ & $T_{\text{co}} = 592.5 \text{ K}$, ASSUME LINEAR DROP $\Rightarrow$
 RADIAL MIDPOINT OF CLADDING IS $\frac{T_{\text{ci}} + T_{\text{co}}}{2} = \frac{618.75 + 592.5 \text{ K}}{2}$
 $\Rightarrow$ MIDPOINT (R) OF CLADDING IS 605.63 K

FOR $r = 0.1 \text{ cm}$ INTO FUEL:

$$T(r) = \frac{Q(R_f^2 - r^2)}{4k_f} + T_s = \frac{350(0.45^2 - 0.1^2)}{4(0.04)} + 861.06 \text{ K}$$

* * * Scribble
 wrong,
 come
 back

$$T_f(0.1 \text{ cm}) = 1325.9 \text{ K}$$

$$T_f(0.1 \text{ cm}) = 1282.15 \text{ K}$$

3. CALCULATE LHR @ $Z=2.1m$ AND T_{cool} @ $2.1m$

GIVEN $L_{rod} = 3.6m$

$LHR_0 = 400 W/cm$

$\gamma = 1.25$

ASSUME: $C_p = 4200 J/kg \cdot K$

$\dot{m} = 0.1 kg/s \cdot rod$

$T_{in} = 580K$

*WATCH
UNITS*

$$Z_0 = \frac{L}{2} = \frac{3.6m}{2} = 1.8m$$

$$LHR_Z = LHR_0 \cos\left(\frac{\pi}{2\gamma}\left(\frac{Z}{Z_0} - 1\right)\right)$$

$$LHR_{(2.1m)} = 400 W/cm \cdot \cos\left(\frac{\pi}{2(1.25)}\left(\frac{2.1}{1.8} - 1\right)\right)$$

$$LHR_{(2.1m)} = 391.26 W/cm$$

$$T_{cool}(2.1m) = \frac{2\gamma}{\pi} \cdot \frac{Z_0 \cdot LHR_0}{\dot{m} \cdot C_p} \left[\sin\frac{\pi}{2\gamma} + \sin\left[\frac{\pi}{2\gamma}\left(\frac{Z}{Z_0} - 1\right)\right] \right] + T_{in}$$

$$T_{cool}(2.1m) = \frac{2(1.25)}{\pi} \cdot \frac{1.8m \cdot 400}{0.1 \cdot 4200} \left[\sin\frac{\pi}{2(1.25)} + \sin\left[\frac{\pi}{2(1.25)}\left(\frac{2.1}{1.8} - 1\right)\right] \right] + 580K$$

$$T_{cool}(2.1m) = 734.40K$$

T_{cool} @ outlet!

4. CALC THERMAL COND. DEGRADATION OF VOZ @ BURN OF
5% FIMA

ASSUME FUEL TEMP $T_{fuel} = 1200K$

$$K_{ox} = \frac{1}{A + BT}$$

$$A = 3.8 + 200(FIMA) = 3.8 + 200(0.05) = 13.8$$

$$B = 0.0217$$

$$K_{ox} = \frac{1}{13.8 + 0.0217(1200K)}$$

$$K_{ox} = 0.025 W/cm \cdot K$$

$$K_{new} = \frac{1}{3.8 + 0.0217(1200K)} = 0.034 W/cm \cdot K$$

$$\Rightarrow \frac{K_{new} - K_{ox}}{K_{new}} \times 100 = \frac{0.034 - 0.025}{0.034} = 26.477\% \text{ REDUCTION}$$

5. TO DRIVE AN ARBITRARY PROFILE FOR TEMPERATURE YOU NEED A CONSTANT THERMAL CONDUCTIVITY, STEADY STATE $\frac{dT}{dx} = 0$, ONE-DIMENSION FLOW, NO INTERNAL HEAT GENERATION

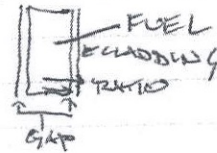
- we used Q for the fuel
- just volume averaged Q

6. FOR ENRICHMENT UF_6 IS USED. THE PROCESS FOR CENTRIFUGAL ENRICHMENT HAS GAS ENTER A HIGH SPEED CENTRIFUGE AND THIS MOVES HEAVIER $U-238$ TO OUTSIDE ALLOWING $U-235$ TO BE SUCKED UP FROM CENTER.

7. CHF IS THE POINT AT WHICH BOILING (USUALLY NUCLEATE BOILING) STOPS BEING EFFECTIVE @ TRANSFERRING HEAT FROM CLADDING TO COOLANT. DNB IS THE RATIO > 1 TO ENSURE MARGIN AWAY FROM HEAT FLUX THRESHOLDS.



8. SMEAR DENSITY IS THE FRACTION OF CLADDING CROSS-SECTION THAT IS OCCUPIED BY FUEL. THE LOWER THE SMEAR DENSITY THE BETTER ENVIRONMENT TO ALLOW FUEL SWELLING + MINIMIZE FUEL/CLADDING INTERFACE



- almost $\frac{R_p}{R_{ic}}$

9. TWO FORMS OF VO_2 DEGRADATION ARE LATTICE DAMAGE FROM RADIATION + BUILD UP OF FISSION GASES.

10. Zr FOR CLADDING IS GOOD BECAUSE IT HAS A LOW CROSS-SECTION FOR NEUTRON ABSORPTION, + RELATIVELY good CORROSION RESISTANCE.

CLADDING IS NEEDED TO PROVIDE STRUCTURAL SUPPORT + ISOLATE FUEL FROM COOLANT.

IVAN DAVIS

NES33

2/9/26

P95

3/4 11. THE FUEL SYSTEM IS THE FUEL PUMP, CANNING, ASSEMBLY, CORE, CONTROL RODS - gyp + coolant

1/4 12. THE FUEL IS A GOOD THERMAL PERFORMER, CAN WITHSTAND CHEMICAL BREAKDOWN, + CAN MAINTAIN STRUCTURE (NO OR LIMITED SWELLING) - mech, not really

5/4 13. THE THREE WAYS ARE FINITE VOLUME, FINITE DIFF, AND FINITE PARTICLE. elements

FINITE DIFFERENCE HAS A HIGH COMP. EFF W/ LOW MEMORY NEEDED W/ GOOD ACCURACY.

4/5 14. UO₂ IS POSITIVE IN THAT IT IS A ^{much...} WELL KNOWN + UNDERSTOOD FUEL SOURCE W/ A RELATIVELY HIGH MELTING POINT.

UO₂ DRAWBACKS ARE IT IS CERAMIC + CAN FRACTURE + HAS A LOW THERMAL CONDUCTIVITY.

4/4 15. EXPLICIT COMPUTE THE NEXT TIME STEP FROM KNOWN VALUES

IMPLICIT SOME USING UNKNOWN FUTURE VALUES

